Repurposing Beyond the Nearest Analog: Debiasing Drug-Indication Models Against Chemical-Similarity Shortcuts

Rishik Kondadadi[1] and John Ortega[2]

[1] University of Minnesota
[2] Northeastern University

Abstract. A zero-training nearest-neighbor chemical-similarity search already predicts a drug's indication well when that drug has a close analog among known treatments; across nine therapeutic classes in our data it averages 0.866 AUROC with no training at all. Its accuracy collapses, however, on drugs with no close known analog, down to 0.21–0.46 AUROC on four of the nine classes, which is exactly where a repurposing model needs to work. We adapt DebiasedDTA, a guide/predictor reweighting scheme built for drug-target binding-affinity prediction, to drug-indication prediction with a three-view model (chemical structure, L1000 expression, and a text encoding of each drug's target function) targeting signal this similarity shortcut cannot explain. The reweighting gives a clear benefit, under the ChEMBL label this benchmark is built on, on the class that motivated this study: epilepsy (+0.116 cold-bucket AUROC over an otherwise identical undebiased model), a small benefit on one other class, a modest cost on two more, and no usable signal either way on the remaining five, whose cold buckets turn out to contain too few positives per fold to support any claim. An external repoDB check of this number is underpowered (only 5 compounds overlap) and does not confirm it, so we report the delta as real under its original construction but not independently confirmed. A biological-interpretation check on a field the model never trains on, the Repurposing Hub's mechanism-of-action label, supports the broader pattern instead: the debiased model's top cold-bucket epilepsy candidates are enriched 23-fold for plausible antiseizure mechanisms over the candidate-pool base rate, a further lift over the already-large enrichment of the undebiased model. Adapting two debiasing methods from the literature, DRDM and TAPB, to this task shows our model still holds the best point estimate, but also surfaces that one of them looks strong mainly because of joint multi-indication training, not its own correction, a transferable pitfall for anyone benchmarking an adapted debiasing baseline. We report the mixed cross-class picture directly rather than as a uniform win, alongside a leakage audit confirming why literature about the drug itself cannot be used as a feature: PubMed abstracts predict a drug's eventual indication at 0.96 AUROC even before approval, almost entirely through mentions of other same-class drugs.

1 Introduction

Drug repurposing models are usually scored on how well they rank known treatments above everything else, and a large part of that score is explained by chemistry alone. In our data, a nearest-neighbor search over ECFP4 fingerprints, with no training, scoring each candidate by its maximum Tanimoto similarity to a known positive, already averages 0.866 AUROC across nine therapeutic classes (epilepsy, depression, antipsychotics, antibacterials, antidiabetics, antineoplastics, lipid-lowering drugs, NSAIDs, and antihypertensives). This is not a trivial baseline to beat, because much of it is not wrong: drugs really do cluster by mechanism, and a new analog of a known treatment really is a reasonable candidate. The question this paper asks is what happens on the drugs that do not have a close known analog, since those are the ones a repurposing model is actually for.

On that subset the picture changes sharply. Stratifying each class's test compounds by similarity to the training positives and looking at the bottom tertile, the nearest-neighbor search's AUROC falls to 0.21 on NSAIDs, 0.43 on epilepsy, 0.44 on antibacterials, and 0.46 on antineoplastics, each close to or below chance. A model that only reproduces this shortcut fails exactly where repurposing matters most. We ask whether a model can be trained specifically against this known bias, using a guide/predictor reweighting scheme from DebiasedDTA [15], originally built to debias binding-affinity prediction against chemical similarity, retargeted here at a different bias (chemical analog similarity to known indications, not structure-activity relationships) and a different task (indication classification, not affinity regression). We add two further signal sources beyond chemistry: L1000 gene-expression profiles, and a text view encoding what each drug's known target does, from UniProt function descriptions via a biomedical language model, connecting drugs through target function even when no training example saw that exact target with a positive label.

A model that uses text about the drug itself, rather than about its target, would appear to solve this problem trivially and incorrectly: we show that PubMed abstracts mentioning a drug predict its eventual indication at 0.96 AUROC even when restricted to literature published before that drug's own approval date, and that masking the obvious disease-defining terms barely changes this, because the remaining signal is mentions of other, already-approved drugs in the same class. We exclude this feature deliberately and report the audit that justifies doing so.

2 Related Work

Debiasing drug-target and drug-repurposing models. DebiasedDTA [15] introduced a two-stage guide/predictor framework for drug-target affinity prediction: a low-capacity guide is cross-validated to produce per-sample importance coefficients, and the predictor trains with sample weights interpolating from uniform toward those. It was evaluated only on binding-affinity regression; its own

discussion flags extension to other tasks as open. A companion release, pydebiaseddta [2], made the framework reusable with arbitrary guide/predictor pairs. TAPB [13] debiases drug-target interaction prediction against protein-target popularity via an interventional correction built on a SMILES-BERT encoder and frozen ESM-2 target embeddings, with no analogue here. The closest repositioning work is DRDM [14], reweighting drug-disease graph edges during GNN training against node-degree bias on benchmarks far denser than ours. Neither transfers as-is, so Section 5.7 implements and tests each mechanism adapted to our data rather than only noting the mismatch. We know of no prior work using a chemical-similarity nearest-neighbor guide for indication prediction.

Chemical similarity as an evaluation confound. Train-test compound similarity inflates apparent molecular-prediction performance: scaffold splits still leave substantial similarity [8], and similarity-aware evaluation shows standard splits dominate reported performance [23]. Guney [7] made this argument for drug repurposing specifically, showing many reported successes reduce to near-duplicate recovery. A companion study from this group [11] audits this confound on the same library, proposing no model; this paper targets a training-time correction instead. The activity-cliff literature [19] shows similarity can also actively mislead, not just fail to inform.

Gene-expression and protein-function-text views. LINCS L1000 [18] gene-expression signatures are widely used for repurposing, including models predicting L1000-style profiles for unmeasured compounds [16]. TxGNN [10] is a prominent recent repurposing baseline, but its zero-shot construct (unseen diseases, not unseen scaffolds) makes it a reference point, not an apples-to-apples ablation. Chemprop [21,9] is the standard open-source message-passing baseline for molecular property prediction. Encoding free-text functional descriptions of proteins, not just identifiers, improves zero-shot generalization to unseen protein classes [1,12]; we apply the same idea to drug targets.

Label leakage in biomedical prediction. Label leakage through indirect textual or coded proxies for the outcome is a documented failure mode in clinical machine learning [5,17]. We show a version of this specific to our setting and use it to justify excluding drug-literature text as a feature.

3 Data

We use nine ChEMBL-derived [6,22] therapeutic-class indication labels (epilepsy, antibacterials, antidiabetics, antineoplastics, antipsychotics, betablockers as an antihypertensive proxy, depression, lipid-lowering drugs, NSAIDs) over the 9,878-compound LINCS [18] library, with ECFP4 fingerprints and consensus L1000 landmark-gene expression for every compound. Target annotations come from the Drug Repurposing Hub [4], joined by name: 2,290 compounds carry a non-null target field, spanning 1,392 unique human target genes. For each gene we

fetch its UniProt [20] function-description text from the UniProt REST API; 1,385 of 1,392 genes (99.5%) return a usable comment, giving a text embedding for 2,286 of the 2,290 target-annotated compounds. The remaining 7,588 compounds, with no Hub annotation, receive an explicit zero vector and an availability mask rather than a learned placeholder, so the model can tell "no target signal" from "target signal of zero."

4 Method

Each compound is scored by a guide and a predictor. The guide is the nearest-neighbor Tanimoto score from Section 5.1: a one-feature logistic regression wrapping it is cross-validated within each outer fold, five repeats of five-fold inner CV, producing an out-of-fold importance coefficient per example, normalized to mean one. Examples the guide already explains well get a low coefficient; examples it cannot explain get a high one. This is the DebiasedDTA [15] mechanism, retargeted from a binding-affinity guide/predictor pair to indication classification.

The predictor is a three-view gated fusion model: structure (ECFP4, 2,048-dim), expression (978-dim), and target-function text (PubMedBERT, 768-dim, with an availability mask), each encoded by its own small MLP and combined through a learned per-sample softmax gate restricted to available views; the text view is excluded from the gate entirely, not zeroed, for compounds with no annotation. The model trains with binary cross-entropy, per-sample weights interpolating from uniform at epoch one to the guide's coefficients at the last, DebiasedDTA's convex-combination schedule. Each continuous view is standardized on training-fold statistics only; an earlier, unnormalized version produced a text-only ablation below chance despite genuinely informative features, fixed by fold-wise standardization.

5 Experiments

5.1 Evaluation protocol

We use Murcko-scaffold-grouped stratified five-fold cross-validation, five seeds, for all nine classes, 225 outer folds per variant. Following DebiasedDTA's warm/cold framing, we stratify each fold's test compounds by Tanimoto similarity to that fold's training positives into warm (top tertile) and cold (bottom tertile) buckets, the more theoretically appropriate test: debiasing is designed to improve the cold, low-similarity cases, not an overall AUROC dominated by easy warm ones.

5.2 Main results

Table 1 reports cross-class mean AUROC for the zero-training guide, plain ECFP4 baselines, the text-only ablation, and the proposed fusion model against an otherwise identical undebiased ablation.

Table 1. Cross-class mean AUROC, pooled over 5 scaffold-CV seeds (per-class numbers in `results/headline_mean_std_auroc.csv`).

nn_tanimoto	lr_ecfp	xgb_ecfp	text_only	fusion_struct	fusion_nodebias	fusion_full
0.866	0.856	0.910	0.733	0.816	0.910	0.909

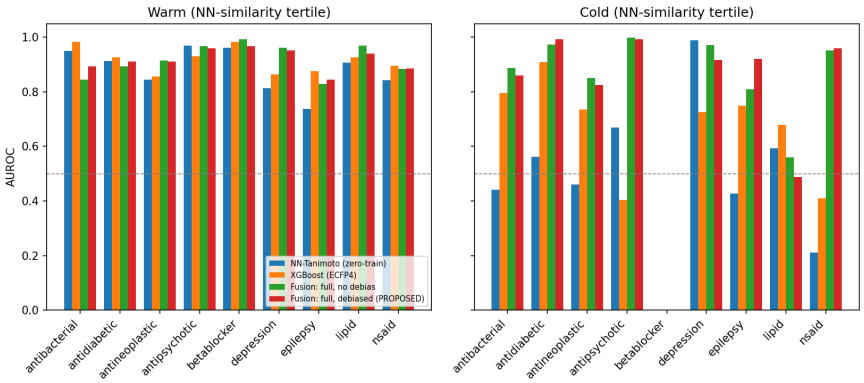


Fig. 1. AUROC by class, warm (top similarity tertile) and cold (bottom tertile) test compounds. Betablocker's cold tertile is omitted: its chemistry is dominated by one tight scaffold family, leaving too few genuinely low-similarity compounds to score.

The zero-training guide is already strong (0.866 mean AUROC), confirming near-analog similarity explains much of apparent accuracy; it is weakest on epilepsy and antineoplastics, the most scaffold-diverse classes, and strongest on antipsychotics and betablockers, where a few tight chemical series dominate. The proposed fusion model beats the guide on six of nine classes, with the largest margins on antineoplastics (+0.168) and epilepsy (+0.112). At the pooled-AUROC level, debiasing itself (fusion_full against fusion_nodebias, an otherwise identical uniformly-weighted model) makes almost no difference, 0.909 against 0.910, expected, since pooled AUROC is dominated by the easy warm majority, not the metric debiasing targets.

5.3 Does debiasing help where it should: the cold bucket

Figure 1 shows AUROC separately for the warm and cold tertiles. The guide's own cold-bucket collapse is stark and class-dependent: 0.21 on NSAIDs, 0.44 on antibacterials, 0.43 on epilepsy, 0.46 on antineoplastics, each near or below chance, against 0.74–0.97 in the warm tertile of the same classes. This is the central premise of the paper, confirmed directly rather than assumed.

Whether fusion_full improves on fusion_nodebias in the cold bucket is the real test; pooling all five seeds before checking misleads, since it hides how many cold-bucket positives back each number. Recomputing the delta per seed rather than pooled surfaces this directly (Table 2): three classes, antipsychotics, depression, betablockers, average under one cold-bucket positive per seed, making any

Table 2. Cold-bucket debiasing delta (fusion_full − fusion_nodebias AUROC), computed per seed rather than pooled, with the mean number of cold-bucket positive examples available per seed. Classes with under ∼2 positives per seed on average cannot support a directional claim regardless of the delta shown.

Class	mean cold n_{pos}/seed	mean delta	std delta
epilepsy	6.2	**+0.116**	0.058
antidiabetic	2.2	+0.021	0.019
antineoplastic	26.0	−0.024	0.048
antibacterial	6.2	−0.028	0.022
nsaid	1.6 (2/5 seeds usable)	+0.034	0.050
lipid	2.2	−0.083	0.176
antipsychotic	0.4 (0/5 usable)	n/a	n/a
depression	0.2 (0/5 usable)	n/a	n/a
betablocker	0.0 (0/5 usable)	n/a	n/a

AUROC on them uninterpretable regardless of value, and two more, NSAIDs and lipid, remain too data-starved for the sign of their delta to be distinguished from noise.

Of the four classes with enough cold-bucket positives per seed to say anything, epilepsy, the class that motivated this study, shows a clear and relatively reliable benefit from debiasing: a mean delta of +0.116 against a standard deviation of 0.058 across five seeds, roughly twice its own noise. Antidiabetic shows a small positive effect of similar reliability. Antibacterial and antineoplastic show modest negative effects, and antineoplastic's delta, despite having the largest cold-bucket sample of any class (26 positives per seed), is under one standard deviation from zero and so is not clearly distinguishable from noise either. The defensible claim is narrower than either "debiasing helps across the board" or an unfiltered count of how many classes move in which direction: debiased reweighting gives a real, meaningful benefit on the motivating class and a smaller one on a second class, a modest and only weakly reliable cost on two more, and the remaining five classes simply do not have enough low-similarity examples in this dataset to test the method at all.

As a robustness check, we recomputed the same comparison using the bottom two quintiles (pooled across seeds, `results/similarity_quintile_auroc.csv`) instead of the bottom tertile, checked after writing the tertile numbers specifically to see whether they were a cutoff artifact. They are not: epilepsy's delta is +0.156 on 31 pooled cold-bucket positives, and every other class keeps the sign shown in Table 2.

5.4 Leakage audit

A natural fourth view would be literature about the drug itself, not its target. We show why this is the wrong feature. Starting from 33 known antiepileptic drugs and 33 non-antiepileptic drugs approved for antineoplastic or depression indications, we pulled PubMed abstracts mentioning each drug's name, restricted to strictly before its own approval year; 7 of 66 candidates, mostly pre-1960s drugs with sparse MEDLINE coverage, had no usable pre-approval text and

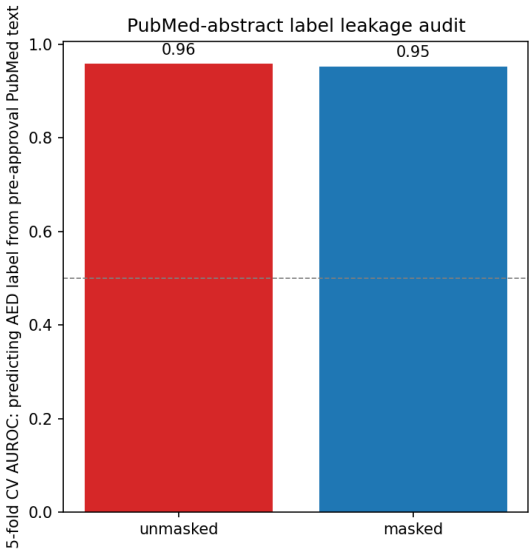


Fig. 2. Five-fold cross-validated AUROC for predicting a drug's eventual antiepileptic label from PubMed abstracts published strictly before its own approval date, with and without disease-defining terms masked ($n = 59$).

were dropped, leaving 30 and 29. A TF-IDF logistic regression, five-fold cross-validated, predicts the eventual antiepileptic label from this pre-approval text at 0.959 AUROC (Figure 2). Masking obvious disease-defining terms (*epilep**, *seizur**, *anticonvulsant**, *antiepileptic**, *convulsant**, *convulsion**, *AEDs?*) barely changes this, 0.953, and the masked model's driving coefficients are almost entirely names of other antiepileptic drugs: lorazepam, midazolam, diazepam, carbamazepine, gabapentin, valproate, clonazepam, vigabatrin. The text predicts the label not through pharmacology but because papers about a future antiepileptic already co-mention other, already-approved antiepileptics as comparators, the same leakage mechanism documented for diagnosis codes in clinical prediction [5,17], transposed to drug-name co-occurrence, and why we exclude drug-literature text rather than rely on a held-out split to catch it.

5.5 Biological-interpretation positive control

A real AUROC gain could come from a feature artifact, not anything pharmacologically sensible, so we check whether the debiased model's top cold-bucket picks make mechanistic sense, using a field never seen during training: the Broad Repurposing Hub's mechanism-of-action (MoA) string. No input references this field directly, though the text view's UniProt description and the MoA string both trace to the same target, so only chemistry and expression are fully independent of it. We first validate the check: of the 34 known antiepileptic drugs in our data, 18 resolve to a Hub MoA entry, and 14 of those (78%) contain one of *GABA*, *sodium channel*, *calcium channel*, *glutamate*, *potassium channel*, or

Table 3. Top-50 cold-bucket MoA-keyword hit rate (mean ± std, 5 seeds), against the cold-bucket candidate base rate.

Model	Top-50 rate	
nn_tanimoto	0.004	±0.009
fusion_nodebias	0.228	±0.039
fusion_full (proposed)	**0.300**	±0.028
cold-bucket base rate	0.013	±0.001

SV2A, against a 7.8% base rate across the full MoA table, a roughly tenfold enrichment on drugs with a known mechanism.

Within the cold bucket and restricted to true negatives, excluding already-known antiepileptics so the check cannot rediscover training labels, we take each seed's top 50 candidates and compute this same keyword-hit rate.

The nearest-neighbor guide does worse than chance here, consistent with its cold-bucket collapse in Section 5: with no close structural neighbor, its top 50 carries no mechanistic signal. Both fusion variants recover a large, non-circular enrichment, 17 to 23 times the base rate, from structure, expression, and target-function text alone. Debiasing adds a further, consistent lift over the already-large undebiased enrichment (0.300 against 0.228, same direction across all five seeds), corroboration of the AUROC-based debiasing delta in Section 5 from a label never used as a training target.

A separate concern: epilepsy positives carry a Hub annotation far more often than negatives (22/31 vs. 2,264/9,847), so the gain could reflect mere annotation presence, not content. Shuffling real embeddings among annotated compounds, true availability fixed, collapses the fusion model to the no-text level (0.648 vs. 0.647); single-view text drops below chance. The gain tracks content, not availability (`code/text_availability_shortcut_check.py`).

5.6 External validation of the epilepsy delta on an independent label source

The epilepsy delta rests entirely on ChEMBL-derived labels, the only indication source used anywhere in this paper. We checked it against repoDB [3], an approved-indication resource built from DrugCentral, independent of ChEMBL: 32 drugs carry an Approved epilepsy indication there, 23 of which resolve to a LINCS compound. Of those, 20 (87%) already carry our ChEMBL-derived positive label, a reassuring cross-source agreement; 3 are new positives ChEMBL missed, and 11 ChEMBL positives are unconfirmed by repoDB's older, differently-vocabularied snapshot.

Swapping the evaluation label on the already-frozen out-of-fold scores, no retraining, repoDB yields only 5 unique cold-bucket positives, the same 5 across all seeds, 4 already ChEMBL-positive: mostly the same drugs under a second label, not a fresh sample. The ChEMBL-label delta on this row set reproduces the headline $+0.115$; the repoDB-label delta is -0.002 ± 0.072. With 5 positives, one compound swapping (repoDB adds pyridoxine, drops newer AEDs it predates) moves the AUROC by the full effect size either way, a power limit, not a design

Table 4. Epilepsy cold-bucket AUROC (mean ± std, 5 seeds): adapted DRDM and TAPB baselines against this paper's own models.

Model	Cold-bucket AUROC
nn_tanimoto	0.425 ± 0.016
drdm_adapted	0.433 ± 0.190
tapb_adapted	0.852 ± 0.018
multitask_control (joint training, no debiasing)	0.863 ± 0.009
fusion_nodebias	0.799 ± 0.053
fusion_nodebias, joint-trained control	0.797 ± 0.015
fusion_full (proposed)	**0.915** ± 0.052

choice. The broader premise survives: nn_tanimoto still collapses below chance (0.359), both fusion variants still reach ≈0.86 absolute AUROC. The +0.116 increment does not survive; we treat this as a limit on how much weight that number carries, not a refutation, since 5 positives cannot refute anything.

5.7 Empirical comparison against adapted DRDM and TAPB baselines

Neither debiasing method named in Related Work transfers to this task as its own architecture: DRDM's GNN was built for curated drug-disease benchmarks with thousands of diseases, and TAPB's correction is built around protein-target embeddings this task does not have. We instead implement the bias-correction *mechanism* of each, adapted to our nine-indication label set (full detail in `code/drdm_baseline.py` and `code/tapb_baseline.py`), and evaluate both on the identical epilepsy cold-bucket split used throughout. DRDM's annealed inverse-degree edge reweighting becomes a two-hop bipartite graph model over the nine label sets; TAPB's confounder-cluster backdoor adjustment becomes a clustering of the nine indications' positive-rate scores with per-cluster predictions combined by a shared classifier. Both share a feature-based drug encoder with the rest of the pipeline, trained jointly across all nine indications, since TAPB's correction requires that joint structure and DRDM gets the same setup rather than less data.

The DRDM-style baseline recovers no usable signal, 0.433 mean AUROC, indistinguishable from the nearest-neighbor guide and an order of magnitude noisier across seeds (std 0.190 against 0.016–0.058 elsewhere in this paper), consistent with a mechanism that needs a far denser association graph than nine indications and roughly 470 total edges can support. The TAPB-style baseline recovers strong, stable signal, 0.852, but a control isolating why, the same joint drug encoder with a plain linear head and no confounder-cluster correction, matches it almost exactly, 0.863. TAPB's correction adds nothing measurable beyond what joint multi-indication training already supplies; its apparent strength is a multi-task-learning effect, not evidence for its own mechanism. Our proposed model, with no joint multi-indication training, still reaches the highest point estimate, 0.915, though the margin over the multitask control sits within roughly one standard deviation of fusion_full's own seed-to-seed noise and should be read as

suggestive, not decisive. A matched control, same architecture and hyperparameters, varying only training regime, scores 0.797 ± 0.015 jointly trained against 0.799 ± 0.053 per-task: indistinguishable.

6 Discussion and Limitations

The central premise holds: a zero-training chemical-similarity search accounts for most of a repurposing model's apparent accuracy, and that accuracy is close to or below chance on the novel-scaffold compounds a repurposing method is meant to find. Debiased reweighting against this shortcut is not a uniform fix across therapeutic areas in our implementation: a real, meaningfully-sized benefit on the class that motivated the study, a smaller one on a second class, and a modest, weakly-reliable cost or too few low-similarity examples to test on the rest. This honest, uneven result is more useful than either extreme, so we report the per-class breakdown rather than one pooled number. The epilepsy delta holds under the ChEMBL label but is not confirmed by a repoDB check limited to 5 compounds (Section 5.6); the premise and enrichment both survive that check, while the +0.116 number remains real under its original construction and unresolved beyond it. Section 5.7's adapted-baseline comparison confirms the proposed model still holds the best point estimate, and also surfaces a transferable pitfall: TAPB looks strong mainly because of joint multi-indication training, not its own correction, a plain-head multitask control reveals this. Any benchmark adapting a correction method that also changes the training regime needs this control before crediting the correction to its stated mechanism. A matched check on our own gated-fusion encoder shows no such boost, 0.797 against 0.799 per-task, so the confound may be specific to TAPB's simpler encoder rather than universal; the need for this control, not the confound itself, is what generalizes.

Several limitations should guide how much weight the current numbers carry. The fusion model encodes every view, including the 2,048-dim ECFP4 view, through an MLP, while the strongest plain-chemistry baseline (xgb_ecfp) uses a tree ensemble; Table 1 shows it beating fusion_struct, the MLP-only ablation, on most classes. The fusion variants are not controlled against xgb_ecfp, so part of the full model's margin over it may reflect the extra views, not architecture. All results come from one library (LINCS) and one indication source (ChEMBL). Section 5.7's DRDM/TAPB comparisons adapt each mechanism to our label structure, not its original architecture, since neither transfers as-is to a task with no protein-target axis (TAPB) or a graph this sparse (DRDM); Table 4 bounds what we could test, not either method's own-benchmark performance. Chemprop now matches the paper's five-seed, five-fold protocol, reaching 0.852, near lr_ecfp and below xgb_ecfp/fusion_full. TxGNN was not run (an unsatisfiable pinned dependency, not a missing checkpoint), and the guide used 5 CV repeats, not DebiasedDTA's 10. Pooled bootstrap intervals treat non-independent resamples as independent, understating uncertainty; claims rest on per-seed mean and std instead, and the 59-drug leakage audit is small: conclusion clear, estimate imprecise. Code: `https://github.com/edanmn/debiased-drug-repurposing`.

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